

Random Variables

Statistics I • Section 4

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Outline

- 4.1 Random variables and distributions
- 4.2 Expectation and variance
- 4.3 Moments and the moment generating function
- 4.4 The Markov and Chebyshev inequalities
- 4.5 Joint, marginal and conditional distributions
- 4.6 Expectation, independence, covariance and correlation
- 4.7 Transformations of random variables

Random Variables and Distributions

Random variables

Definition 4.1 (Random variable)

A *random variable* (隨機變數) (RV) $X : S \rightarrow \mathbb{R}$ assigns a real number to each outcome.

- It turns outcomes into numbers. Then we can average them, measure spread, and combine them.
- Capital X is the variable, a function on S . Small x is one fixed number. So “ $X = x$ ” is an event.
- Every statistic of Section 7 is a random variable.

Example 4.2

Example 4.2

Toss a fair coin twice, $S = \{HH, HT, TH, TT\}$. Let X be the number of heads. Let $Y = 1$ if the first toss is a head, else 0. Let Z be the payoff of a bet that wins \$3 on HH and loses \$1 otherwise. (a) Tabulate $X(s), Y(s), Z(s)$. (b) Write $\{X = 1\}$ as a subset of S and find its probability. (c) Find the probability that $Z = -1$.

Real random variables

- A stock's return tomorrow: the outcome is tomorrow's state of the market. X is the return it produces.
- The number of defaults among ten loans: values $0, 1, \dots, 10$.
- The waiting time for the next customer: values in $[0, \infty)$.
- In each case there is a random experiment and a rule that gives a number. The sample space need not be listed.

A function of a random variable

Proposition 4.3 (A function of a random variable)

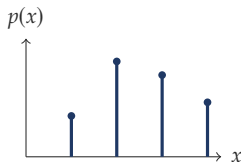
Let X be a random variable and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a function. Then $Y = g(X)$ is also a random variable.

- $Y(s) = g(X(s))$ assigns a real number to each outcome. That is all Definition 4.1 asks for.
- So X^2 , $2X + 3$ and e^X are random variables.
- In Example 4.2, X^2 takes the values 4, 1, 1, 0 on HH, HT, TH, TT .

Discrete random variables

Definition 4.4 (Discrete RV; pmf)

X is *discrete* if its values form a finite or countable set. Its *probability mass function* (機率質量函數) (pmf) is $p(x) = P(X = x)$, with $p(x) \geq 0$ and $\sum_x p(x) = 1$.



- The probability mass function is how a discrete distribution is written down.

Example 4.5

Example 4.5

X = number of heads in three fair tosses. Find the pmf of X .

Example 4.6

Example 4.6

(a) Find the value of c that makes p_X a pmf:

$$p_X(x) = c(2x - 1), \quad x = 1, 2, \dots, 10.$$

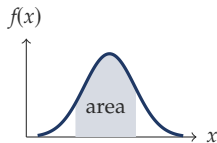
(b) Let $a > 0$ be a constant. Show that p_Y has the properties of a pmf:

$$p_Y(y) = \frac{e^{-a} a^y}{y!}, \quad y = 0, 1, 2, \dots$$

Continuous random variables

Definition 4.7 (Continuous RV; pdf)

X is *continuous* if $P(a \leq X \leq b) = \int_a^b f(x) dx$ for all $a \leq b$, with a *probability density function* (機率密度函數) $f \geq 0$ and $\int_{-\infty}^{\infty} f(x) dx = 1$. The probability of any single value is then zero.



- The probability density function is not itself a probability. It can exceed 1. Height at a point tells us nothing on its own.

Example 4.8

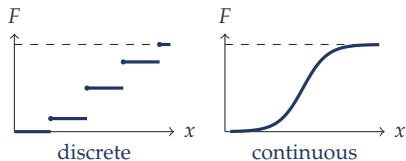
Example 4.8

(a) $f(x) = 2x$ on $[0, 1]$, else 0. Verify it is a density and compute $P(\frac{1}{2} < X < 1)$. (b) $f(x) = c x(1 - x)$ on $[0, 1]$, else 0. Find the value of c that makes f a density. Then compute $P(X > \frac{3}{4})$, and find the height of f at $x = \frac{1}{2}$.

The distribution function

Definition 4.9 (CDF)

The *cumulative distribution function* (累積分配函數) (CDF) is $F(x) = P(X \leq x)$. It is nondecreasing and right-continuous, with $F(-\infty) = 0$, $F(+\infty) = 1$. For continuous X , $F' = f$ wherever f is continuous.



- It accumulates probability from the left: a staircase for a discrete variable, a smooth curve for a continuous one.

Probabilities from the CDF

Proposition 4.10

For any random variable and any $a < b$, $P(a < X \leq b) = F(b) - F(a)$. For a discrete variable with finitely many values, write $F(x^-)$ for the value of F just to the left of x . Then $p(x) = F(x) - F(x^-)$, the jump of F at x .

- So a table of F answers every interval question. Sections 5 and 6 give such tables.

A third case: mixed variables

- A variable can be part jump, part density. We call it *mixed*.
- Example: a payout that is 0 with probability 0.7, and otherwise spread over $(0, 1)$.
- Its CDF jumps at 0, then rises continuously.
- We handle it by combining a sum for the jumps with an integral for the density. Example 4.15 does exactly that.

Example 4.11

Example 4.11

(a) For $X = \text{heads in three tosses}$ (Example 4.5), find the CDF and use it to compute $P(X \geq 2)$. (b) For $f(x) = 2x$ on $[0, 1]$ (Example 4.8), find the CDF and use it to compute $P(X \leq \frac{1}{2})$ and $P(\frac{1}{2} < X < 1)$.

Example 4.12

Example 4.12

X takes 1, 2, 3, 4 with $p = (0.1, 0.2, 0.3, 0.4)$. Find $P(X < 2)$, $P(X \leq 2)$, $P(X \geq 2)$ and $P(1 \leq X < 3)$. Which of the four equals $F(2)$?

Example 4.13

Example 4.13

A discrete variable has $F(x) = 0$ for $x < 1$; 0.15 on $[1, 2)$; 0.50 on $[2, 3)$; 0.80 on $[3, 4)$; 1 for $x \geq 4$ (the staircase on the CDF frame). Recover the pmf, and find $P(X > 2)$.

Example 4.14

Example 4.14

$f(x) = x$ on $[0, 1]$, $f(x) = 2 - x$ on $(1, 2]$, else 0. (a) Verify that f is a density. (b) Find $F(x)$ on each piece. (c) Compute $P(0.5 < X < 1.5)$.

Example 4.15

Example 4.15

A payout X has the distribution function

$$F(x) = \begin{cases} 0, & x < 0, \\ 0.2, & 0 \leq x < 1, \\ 0.2 + 0.3(x - 1), & 1 \leq x < 2, \\ 0.7, & 2 \leq x < 3, \\ 0.7 + 0.2(x - 3), & 3 \leq x < 4, \\ 1, & x \geq 4. \end{cases}$$

(a) Sketch F . (b) Find $P(X = 0)$, $P(X = 2)$ and $P(X = 4)$. (c) Find the density of X on $(1, 2)$ and on $(3, 4)$. (d) Find $P(X \leq 1.5)$ and $P(1.5 < X \leq 3.5)$. (e) Find $P(2 \leq X < 4)$. (f) Check that the point masses and the density together carry probability 1.



Expectation and Variance

A convention for Section 4

- Throughout Section 4, every expectation, variance and covariance that appears is assumed finite.
- Every variance or standard deviation in a denominator is also assumed positive.

Expectation

Definition 4.16 (Expectation)

The *expectation* (期望值) is $\mathbb{E}(X) = \sum_x x p(x)$ (discrete), or $\int_{-\infty}^{\infty} x f(x) dx$ (continuous).

- It is the value that the average of many draws settles down to.

Example 4.17

Example 4.17

(a) Roll a fair die and let X be the face shown. Find $\mathbb{E}(X)$. (b) Let X have the pmf $p_X(x) = \frac{2x-1}{100}$ for $x = 1, 2, \dots, 10$, as in Example 4.6. Find $\mathbb{E}(X)$.

Example 4.18

Example 4.18

(a) Let $f(x) = 2x$ on $[0, 1]$, else 0 (Example 4.8). Find $\mathbb{E}(X)$. (b) Let $f(x) = 3/x^4$ for $x \geq 1$, else 0. Verify that f is a density and find $\mathbb{E}(X)$.

Example 4.19

Example 4.19

A fair coin is tossed until the first head appears. If the first head appears on toss k , the game pays $X = 2^k$ dollars. (a) Find the pmf of X and check that it sums to 1. (b) Find $\mathbb{E}(X)$. (c) Would you pay \$20 to play this game? Compare your answer with (b).

Expectation of a function

Proposition 4.20 (Expectation of a function of X)

For any function g , $\mathbb{E}[g(X)] = \sum_x g(x) p(x)$ (discrete), or $\int_{-\infty}^{\infty} g(x) f(x) dx$ (continuous).

- We weight $g(x)$ by the same probabilities as x .
- We do not need the distribution of $g(X)$ first.
- It is often called the “law of the unconscious statistician”.

Example 4.21

Example 4.21

A person values x dollars by the function $u(x) = \sqrt{x}$. The number $u(x)$ is called the *utility* of x dollars. A gamble pays $X = 100$ or $X = 0$ dollars, each with probability $\frac{1}{2}$. (a) Find $\mathbb{E}(X)$. (b) Find the expected utility $\mathbb{E}[u(X)]$. (c) Compare $\mathbb{E}[u(X)]$ with $u(\mathbb{E}(X))$. (d) Find the sure amount c with $u(c) = \mathbb{E}[u(X)]$. Would this person give up the gamble for a sure \$30?

Example 4.22

Example 4.22

Return to the game of Example 4.19, where $X = 2^k$ with probability $(\frac{1}{2})^k$ for $k = 1, 2, 3, \dots$. A player values x dollars by $u(x) = \ln x$. (a) Show that $\sum_{k=1}^{\infty} k(\frac{1}{2})^k = 2$. (b) Find the expected utility $\mathbb{E}[u(X)]$. (c) Find the sure amount c with $u(c) = \mathbb{E}[u(X)]$, and compare with Example 4.19.

Expectation of a linear transformation

Proposition 4.23

For constants a and b , $\mathbb{E}(aX + b) = a \mathbb{E}(X) + b$. More generally, for functions g and h , $\mathbb{E}[a g(X) + b h(X)] = a \mathbb{E}[g(X)] + b \mathbb{E}[h(X)]$.

- A change of units changes the mean in the same way.
- A sum can be taken apart term by term.

Example 4.24

Example 4.24

(a) A game costs \$10 to enter and pays $4X$ dollars, where X is the face of a fair die. Find the mean of the profit $Y = 4X - 10$. (b) Let X have the density $f(x) = 3/x^4$ on $[1, \infty)$ of Example 4.18(b). Find $\mathbb{E}(X^2)$, and then $\mathbb{E}(2X^2 - 4X + 1)$.

$\mathbb{E}[g(X)]$ is not $g(\mathbb{E}(X))$

- Proposition 4.23 is special to linear functions.
- For other functions the two numbers differ in general:

$$\mathbb{E}(X^2) \neq (\mathbb{E}(X))^2, \quad \mathbb{E}\left(\frac{1}{X}\right) \neq \frac{1}{\mathbb{E}(X)}, \quad \mathbb{E}(e^X) \neq e^{\mathbb{E}(X)}.$$

- Example 4.21(c) already showed this for the square root.
- Always compute $\mathbb{E}[g(X)]$ from Proposition 4.20.

Example 4.25

Example 4.25

X takes 1, 2, 4 with probabilities $\frac{1}{2}, \frac{1}{4}, \frac{1}{4}$. Compare $\mathbb{E}(X^2)$ with $(\mathbb{E}(X))^2$, $\mathbb{E}(1/X)$ with $1/\mathbb{E}(X)$, and $\mathbb{E}(e^X)$ with $e^{\mathbb{E}(X)}$.

Variance

Definition 4.26 (Variance and standard deviation)

With $\mu = \mathbb{E}(X)$, $\text{Var}(X) = \mathbb{E}[(X - \mu)^2]$, and $\sigma_X = \sqrt{\text{Var}(X)}$.

- It is the expectation of a function of X , so Proposition 4.20 computes it.
- The variance is the natural measure of spread about μ .

Computing the variance

Proposition 4.27 (Computing formula)

$$\text{Var}(X) = \mathbb{E}(X^2) - \mu^2.$$

- Usually easier than the definition. It needs one expectation and the mean.

Example 4.28

Example 4.28

For the die of Example 4.17, compute $\text{Var}(X)$ twice: from Definition 4.26, and from Proposition 4.27.

Mean and variance of an indicator

Proposition 4.29

Let A be an event with $P(A) = p$. Its *indicator variable* (指示變數) $\mathbf{1}_A$ equals 1 if A occurs and 0 otherwise. Then $\mathbb{E}(\mathbf{1}_A) = p$ and $\text{Var}(\mathbf{1}_A) = p(1 - p)$.

- So every event gives a random variable. Every probability is also an expectation.
- Y in Example 4.2 is the indicator of “the first toss is a head”.
- The variance is largest when the event is hardest to predict, at $p = \frac{1}{2}$.

Variance of a linear transformation

Proposition 4.30

For constants a and b , $\text{Var}(aX + b) = a^2 \text{Var}(X)$.

- Adding b shifts the distribution and leaves the spread unchanged.
- Multiplying by a stretches every deviation from the mean.

Mean zero, variance one

Corollary 4.31

Let X have mean μ and standard deviation $\sigma > 0$, and set $Z = \frac{X - \mu}{\sigma}$. Then $\mathbb{E}(Z) = 0$ and $\text{Var}(Z) = 1$.

- Z counts how many standard deviations X sits from its mean: the population version of Section 2's z -score.
- Section 6 applies this step to the normal distribution.

Example 4.32

Example 4.32

Let X have the density $f(x) = 6x(1 - x)$ on $[0, 1]$ of Example 4.8(b). (a) Find $\mathbb{E}(X)$ and $\text{Var}(X)$. (b) Find $P(\frac{1}{4} < X < \frac{3}{4})$. (c) Let $Y = 10X - 2$. Find $\mathbb{E}(Y)$ and $\text{Var}(Y)$.

Example 4.33

Example 4.33

X takes the values 0, 1, 2, with $\mathbb{E}(X) = 1.1$ and $\mathbb{E}(X^2) = 1.7$. Find the pmf of X and $\text{Var}(X)$.



Moments and the Moment Generating Function

Raw and central moments

Definition 4.34 (Raw and central moments)

Let $k = 1, 2, 3, \dots$ and $\mu = \mathbb{E}(X)$. The k th *raw moment* (原動差) is $\mu'_k = \mathbb{E}(X^k)$. The k th *central moment* (主動差) is $\mu_k = \mathbb{E}[(X - \mu)^k]$.

- A raw moment measures from 0. A central moment measures from the mean.
- $\mu'_1 = \mu$, $\mu_1 = 0$, $\mu_2 = \sigma^2$.

Central moments from raw moments

Proposition 4.35

With $\mu = \mu'_1$,

$$\mu_2 = \mu'_2 - \mu^2,$$

$$\mu_3 = \mu'_3 - 3\mu \mu'_2 + 2\mu^3,$$

$$\mu_4 = \mu'_4 - 4\mu \mu'_3 + 6\mu^2 \mu'_2 - 3\mu^4.$$

- Expand $(X - \mu)^k$ and take the expectation term by term.
- The first line is the computing formula for the variance.

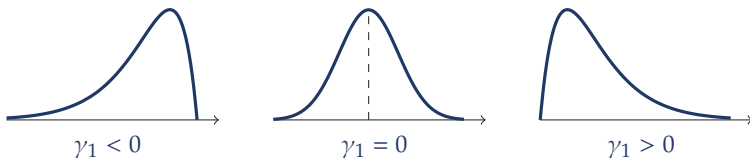
Skewness and kurtosis

Definition 4.36 (Skewness and kurtosis)

Suppose $0 < \sigma^2 < \infty$ and $\mathbb{E}|X|^4 < \infty$. The skewness is $\gamma_1 = \frac{\mathbb{E}[(X-\mu)^3]}{\sigma^3}$. The kurtosis is $\gamma_2 = \frac{\mathbb{E}[(X-\mu)^4]}{\sigma^4}$. The *excess kurtosis* is $\gamma_2 - 3$.

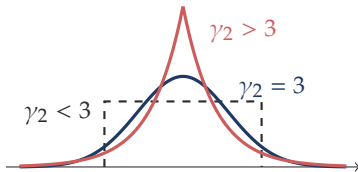
- Dividing by a power of σ removes the units.
- γ_1 is the skewness formula Section 2 promised.

Reading the skewness



- $\gamma_1 > 0$: the large deviations above the mean count for more than the ones below. The right tail is usually longer.
- $\gamma_1 = 0$: the two sides balance. Every symmetric distribution has $\gamma_1 = 0$.
- $\gamma_1 < 0$: the large deviations below the mean count for more than the ones above. The left tail is usually longer.

Reading the kurtosis



- Same mean and same variance in all three. The normal distribution of Section 6 has $\gamma_2 = 3$.
- $\gamma_2 > 3$: heavier tails than normal. Values far from the mean are more likely.
- $\gamma_2 < 3$: lighter tails than normal. Values far from the mean are less likely.
- These readings are guides, not theorems. The kurtosis measures the tails, not the peak.

Example 4.37

Example 4.37

A return X takes the values $-1, 0, 2$ with probabilities $0.2, 0.5, 0.3$. (a) Find the raw moments $\mu'_1, \mu'_2, \mu'_3, \mu'_4$. (b) Use Proposition 4.35 to find μ_2, μ_3 and μ_4 . (c) Find the skewness and the kurtosis, and say what each one tells us.

The moment generating function

Definition 4.38 (MGF)

The *moment generating function* (動差生成函數) (MGF) is $M_X(t) = \mathbb{E}(e^{tX})$, for t near 0 where this is finite.

- It packages all the moments into a single function of t .
- It need not be finite for every t . The MGF *exists* if it is finite on an interval around 0 (Example 4.39).

Example 4.39

Example 4.39

(a) $P(X=1) = p$, $P(X=0) = q = 1 - p$: find $M_X(t)$. (b) X has the density $f(x) = e^{-x}$ for $x \geq 0$, else 0: find $M_X(t)$, and say for which t it is finite.

Where the name comes from

- This result is why the MGF is worth defining at all.

Proposition 4.40

If M_X is finite near 0, then $M_X^{(k)}(0) = \mathbb{E}(X^k)$.

- Differentiate k times and set $t = 0$.

Example 4.41

Example 4.41

(a) Use the MGF of Example 4.39(a) to recover $\mathbb{E}(X)$ and $\text{Var}(X)$, and compare with Proposition 4.29. (b) Let X have the pmf $p(x) = \left(\frac{1}{2}\right)^x$ for $x = 1, 2, 3, \dots$. Find $M_X(t)$, and use it to find $\mathbb{E}(X)$ and $\text{Var}(X)$.

Uniqueness of the MGF

Theorem 4.42

If X and Y share the same MGF on an interval around 0, they have the same distribution.

- This lets us **identify** a distribution from its MGF.
- The proof needs transform inversion, which is beyond this course.

MGF of a linear transformation

Proposition 4.43

For constants a, b , $M_{aX+b}(t) = e^{bt}M_X(at)$.

- The factor e^{bt} is a constant, so it comes out of the expectation.
- Section 6 uses this to pass between a normal variable and the Z of Corollary 4.31.

Example 4.44

Example 4.44

$M_X(t) = 0.2 + 0.5e^t + 0.3e^{3t}$. (a) Read off the distribution of X . (b) Find $\mathbb{E}(X)$ and $\text{Var}(X)$ from the MGF. (c) For $Y = 2X + 1$, find $M_Y(t)$, and from it $\mathbb{E}(Y)$ and $\text{Var}(Y)$.

The Markov and Chebyshev Inequalities

Markov's inequality

Theorem 4.45 (Markov)

Markov's inequality (馬可夫不等式): if $X \geq 0$ and $a > 0$, then $P(X \geq a) \leq \frac{\mathbb{E}(X)}{a}$.

- It bounds a nonnegative variable using only its **mean**.
- No distribution is assumed.

Chebyshev's inequality

Theorem 4.46 (Chebyshev)

Chebyshev's inequality (柴比雪夫不等式): for mean μ , finite variance $\sigma^2 > 0$ and $k > 0$, $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$.

- It follows by applying Markov to $(X - \mu)^2$, and it uses the **variance**.
- Section 2 stated it. We **prove** it here. It returns in Section 7 for the Law of Large Numbers.

Example 4.47

Example 4.47

X = heads in three fair tosses, so $\mathbb{E}(X) = \frac{3}{2}$, $\text{Var}(X) = \frac{3}{4}$. Compare Markov's bound on $P(X \geq 3)$ and Chebyshev's bound on $P(|X - \mu| \geq 2\sigma)$ with the exact values, and say why they are loose.

Example 4.48

Example 4.48

Scores mean 72, standard deviation 4. (a) We want Chebyshev to guarantee at least 90% of scores within k standard deviations of the mean. How large must k be? (b) Write down that interval.

Example 4.49

Example 4.49

X takes $-1, 0, 1$ with probabilities $\frac{1}{8}, \frac{3}{4}, \frac{1}{8}$. (a) Find μ and σ . (b) Compute $P(|X - \mu| \geq 2\sigma)$ exactly and compare with Chebyshev's bound. (c) What does this say about improving the inequality?



Joint, Marginal and Conditional Distributions

Joint and marginal distributions

Definition 4.50

Discrete: the *joint* pmf is $p(x, y) = P(X = x, Y = y)$, and the *marginal* pmf of X is $p_X(x) = \sum_y p(x, y)$. Continuous: a *joint pdf* $f(x, y) \geq 0$ gives the probability of a region as a double integral. The *marginal pdf* of X is $f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$.

- **Joint distribution** (聯合分配): describes the pair at once.
- **Marginal distribution** (邊際分配): describes one variable on its own.

Conditional distributions

Definition 4.51

For $p_X(x) > 0$, the conditional pmf of Y given $X = x$ is $p_{Y|X}(y | x) = \frac{p(x, y)}{p_X(x)}$. For $f_X(x) > 0$, the conditional pdf is $f_{Y|X}(y | x) = \frac{f(x, y)}{f_X(x)}$. The *conditional expectation* (條件期望值) $\mathbb{E}(Y | X = x)$ and the *conditional variance* $\text{Var}(Y | X = x)$ are the mean and the variance of this distribution.

- This is $P(A | B) = P(A \cap B)/P(B)$ of Section 3, for random variables.
- The *conditional distribution* describes Y in the sub-population where $X = x$.

Example 4.52

Example 4.52

$p(x, y)$	$Y=0$	$Y=1$	$Y=2$
$X=0$	0.10	0.08	0.02
$X=1$	0.10	0.25	0.15
$X=2$	0.03	0.09	0.18

(a) Find the marginal pmf of X and the marginal pmf of Y . (b) Find $P(X = Y)$ and $P(X < Y)$. (c) Find the conditional pmf of Y given $X = x$, for $x = 0, 1, 2$. (d) Find $\mathbb{E}(Y \mid X = x)$ and $\text{Var}(Y \mid X = x)$ for each x . (e) Find $P(X \geq 1 \mid Y = 2)$. (f) Let $S = X + Y$. Find the pmf of S and $\mathbb{E}(S)$.

The joint CDF

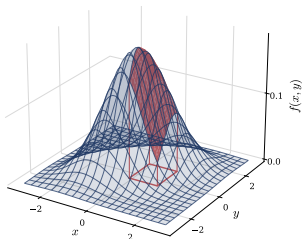
Definition 4.53

The *joint cumulative distribution function* (聯合累積分配函數) (joint CDF) is $F(x, y) = P(X \leq x, Y \leq y)$. For continuous X, Y , $F(x, y) = \int_{-\infty}^x \int_{-\infty}^y f(s, t) dt ds$.

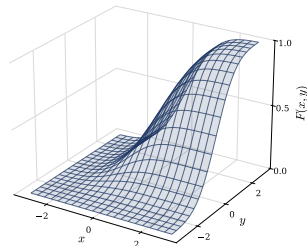
- $F(x, y)$ is the probability of the region to the left of x and below y .
- With one variable, F gave the probability of an interval. With two, it gives the probability of a rectangle.

A joint pdf and its joint CDF

joint pdf: a surface



joint CDF: rises from 0 to 1

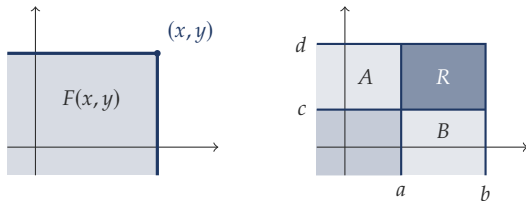


- Left: the joint pdf is a surface. The probability of the rectangle is the volume of the red column above it.
- Right: the joint CDF of the same pair rises from 0 to 1.

Probability of a rectangle

Proposition 4.54

For $a < b$ and $c < d$, $P(a < X \leq b, c < Y \leq d) = F(b, d) - F(a, d) - F(b, c) + F(a, c)$.



- Remove the strips A and B . Their overlap was removed twice, so add it back.

Example 4.55

Example 4.55

X and Y have the joint pdf $f(x, y) = c(x + y)$ for $0 < x < 2$ and $0 < y < 1$, and $f(x, y) = 0$ elsewhere. (a) Find c . (b) Find the marginal pdf of X and the marginal pdf of Y . (c) Find the joint CDF $F(x, y)$ for $0 \leq x \leq 2$ and $0 \leq y \leq 1$. Use it to find $P(1 < X \leq 2, \frac{1}{2} < Y \leq 1)$. (d) Find the conditional pdf of X given $Y = y$, and then $P(\frac{1}{2} < X < \frac{3}{2} \mid Y = \frac{1}{2})$. (e) Find $\mathbb{E}(X \mid Y = \frac{1}{2})$. (f) Find $P(X + Y > 1)$.

Expectation, Independence, Covariance and Correlation

Expectation of a function of two variables

Proposition 4.56

For any function g of two variables, $\mathbb{E}[g(X, Y)] = \sum_x \sum_y g(x, y) p(x, y)$ (discrete), or $\iint g(x, y) f(x, y) dy dx$ (continuous).

- This is Proposition 4.20 for a pair of variables.
- We weight a function of the pair by the joint distribution.

Linearity of expectation

Theorem 4.57

$\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$, for any constants a, b .

- Expectation is **always** linear, whatever the dependence between X and Y .
- No independence is needed.

The joint MGF

Definition 4.58

The *joint moment generating function* (聯合動差生成函數) (joint MGF) of X and Y is $M_{X,Y}(s, t) = \mathbb{E}(e^{sX+tY})$, for (s, t) near $(0, 0)$ where this is finite.

- §4.3 put the moments of one variable into one function. This does the same for a pair.

What the joint MGF gives

Proposition 4.59

Let $M_{X,Y}$ be finite near $(0, 0)$. Then $M_{X,Y}(s, 0) = M_X(s)$ and $M_{X,Y}(0, t) = M_Y(t)$. Differentiate $M_{X,Y}$ once in s and once in t , and then set $s = t = 0$. The result is $\mathbb{E}(XY)$.

- Setting one argument to 0 removes that variable.
- Each derivative brings down one factor: first X , then Y .

Example 4.60

Example 4.60

The joint pmf of (X, Y) is $p(0, 0) = 0.3, p(0, 1) = 0.2, p(1, 0) = 0.1, p(1, 1) = 0.4$. (a) Find the joint MGF $M_{X,Y}(s, t)$. (b) Use it to find $M_X(s)$ and $M_Y(t)$, and then $\mathbb{E}(X)$ and $\mathbb{E}(Y)$. (c) Use it to find $\mathbb{E}(XY)$.

Independence

Definition 4.61 (Independence)

X and Y are independent if $p(x, y) = p_X(x) p_Y(y)$ for all x, y . For continuous X, Y : $f(x, y) = f_X(x) f_Y(y)$.

- If we learn one variable, we learn nothing about the other's distribution.
- This is the strongest way to say the two carry no information about each other.

Functions of independent variables

Proposition 4.62

If X and Y are independent, then for any functions g and h ,
 $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$. This covers $g(x) = x$ and $h(y) = y$ in particular.

- The double sum factors, because $p(x, y) = p_X(x)p_Y(y)$.
- Many later proofs need it: the MGF of a sum (Proposition 4.63), and independence gives zero covariance.
- $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$ alone does not give independence (Proposition 4.75).

MGF of a sum of independent variables

Proposition 4.63

If X and Y are independent, then $M_{X+Y}(t) = M_X(t) M_Y(t)$.

- Sections 5 to 7 all use this. A sum of independent variables is handled through the two separate MGFs.
- Uniqueness then names the sum's distribution.

Example 4.64

Example 4.64

Let X_1, X_2 be independent, each as in Example 4.39(a): 1 with probability p and 0 with probability $q = 1 - p$. Find the MGF of $X_1 + X_2$ and say what distribution it identifies.

Example 4.65

Example 4.65

X and Y are independent. X has the pdf $f_X(x) = 2x$ on $(0, 1)$. Y has the pdf $f_Y(y) = e^{-y}$ on $(0, \infty)$. (a) Write down the joint pdf. (b) Find $P(X > \frac{1}{2}, Y > 1)$. (c) Find $\mathbb{E}(XY)$ and $\mathbb{E}(X^2 e^{-Y})$. (d) Find $P(Y > X)$.

Law of iterated expectations

Theorem 4.66

$$\mathbb{E}[\mathbb{E}(Y \mid X)] = \mathbb{E}(Y).$$

- Average the conditional mean over the distribution of X . The result is the overall mean of Y .

Law of total variance

Theorem 4.67

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y | X)] + \text{Var}[\mathbb{E}(Y | X)].$$

- One piece is the spread that survives inside each value of X . The other is the spread of the group means as X moves.
- Both are nonnegative. So on average, conditioning on X can only remove variability from Y .

Law of total probability

Proposition 4.68

For discrete X, Y , $p_Y(y) = \sum_x p_{Y|X}(y | x) p_X(x)$. For continuous X, Y , $f_Y(y) = \int_{-\infty}^{\infty} f_{Y|X}(y | x) f_X(x) dx$.

- This is the law of total probability of Section 3, for random variables.
- The three laws do the same job. Each rebuilds a fact about Y from the conditional distributions given X .

Example 4.69

Example 4.69

X is equally likely to be 1, 2 or 3. Given $X = x$, the variable Y is equally likely to be any of the values $1, \dots, x$. (a) Find the marginal pmf of Y with the law of total probability. (b) Find $\mathbb{E}(Y \mid X = x)$ for each x , and then $\mathbb{E}(Y)$ with the law of iterated expectations. (c) Find $\text{Var}(Y \mid X = x)$ for each x , and then $\text{Var}(Y)$ with the law of total variance. (d) Check $\mathbb{E}(Y)$ and $\text{Var}(Y)$ from the pmf of Y .

Measuring co-movement

Definition 4.70

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)], \quad \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1].$$

The correlation needs both standard deviations positive and finite.

- The covariance gives the direction of co-movement, and it carries units.
- The correlation rescales it to $[-1, 1]$, so it has no units. Proposition 4.74 proves that bound.

The covariance shortcut

Proposition 4.71

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

- This is the working formula. Use it instead of the definition.

Properties of the covariance

Proposition 4.72

For constants a, b, c, d , $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$. Also $\text{Cov}(X, Y) = \text{Cov}(Y, X)$, $\text{Cov}(X, X) = \text{Var}(X)$, and $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$. If a and c have the same sign, $\rho_{aX+b, cY+d} = \rho_{XY}$; if opposite signs, $\rho_{aX+b, cY+d} = -\rho_{XY}$.

- A change of units, such as percent to decimal, changes the covariance but not ρ_{XY} .

Variance of a linear combination

Theorem 4.73

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).$$

- Two variables that move together do not simply add their spreads. With coefficients of the same sign, positive covariance inflates the variance and negative covariance cancels part of it. With opposite signs it reverses (Example 4.76(c)).

The bound on ρ

Proposition 4.74

$-1 \leq \rho_{XY} \leq 1$. Moreover, $\rho_{XY} = 1$ if and only if $Y = aX + b$ with probability 1 for some $a > 0$ and b . Likewise $\rho_{XY} = -1$ if and only if the same holds with $a < 0$.

- Proof idea: $\text{Var}(X/\sigma_X \pm Y/\sigma_Y) = 2 \pm 2\rho_{XY} \geq 0$.
- So ρ_{XY} measures a **linear** relationship, and ± 1 means an exact line.

Independence gives zero covariance

Proposition 4.75

If X and Y are independent, then $\text{Cov}(X, Y) = 0$. The converse is **false**.

- Zero covariance means no **linear** co-movement, not full independence.
- The counterexample is $Y = X^2$ with X symmetric about 0.

Example 4.76

Example 4.76

Return to the joint pmf of Example 4.52, with marginals $p_X = (0.20, 0.50, 0.30)$ and $p_Y = (0.23, 0.42, 0.35)$ on the values 0, 1, 2. (a) Find $\mathbb{E}(X)$, $\mathbb{E}(Y)$, $\text{Var}(X)$ and $\text{Var}(Y)$. (b) Find $\mathbb{E}(XY)$, $\text{Cov}(X, Y)$ and ρ_{XY} . (c) Find $\text{Var}(X + Y)$ and $\text{Var}(2X - Y)$. (d) Let $U = 2X + 1$ and $V = -3Y + 5$. Find $\text{Cov}(U, V)$ and ρ_{UV} . (e) Are X and Y independent?

Two-asset portfolios

- Weights w_1, w_2 on returns R_1, R_2 : $R_p = w_1 R_1 + w_2 R_2$. Its mean and variance come from Theorems 4.57 and 4.73.
- Take positive weights. Whenever $\rho < 1$, the portfolio's risk falls below the weighted average of the two risks. That is diversification (分散投資).
- Falling below the *smaller* of the two risks needs more: a small enough ρ and the right weights (Example 4.77).

Example 4.77

Example 4.77

Two assets have $\mathbb{E}(R_1) = 8\%$, $\mathbb{E}(R_2) = 12\%$, $\sigma_1 = 10\%$, $\sigma_2 = 20\%$ and $\rho = 0.3$. (a) A portfolio puts 60% in asset 1 and 40% in asset 2. Find its expected return and its standard deviation. Compare with the weighted average of the two standard deviations. (b) Now put weight w on asset 1 and $1 - w$ on asset 2. Write $\text{Var}(R_p)$ as a function $V(w)$. (c) Find the weight w^* that minimizes V , and the resulting σ_p . Is it below the smaller risk, 10%? (d) Repeat (c) with $\rho = -1$.

Many variables at once

Definition 4.78 (i.i.d.)

$X_1, \dots, X_n$ have the joint pmf $p(x_1, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n)$, defined as for two variables. They are *independent* if this joint pmf (or pdf) factors into the product of the n marginals, for every choice of values. They are *independent and identically distributed* (獨立且相同分配) (i.i.d.) if, in addition, all n have the same distribution.

- Sampling with replacement gives i.i.d. draws: each has the population's distribution, and no draw affects another.
- Section 7 builds every sampling distribution on such draws.

Mean and variance of a sum

Proposition 4.79

For $X_1, \dots, X_n$ and constants $a_1, \dots, a_n$,

$$\mathbb{E}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i \mathbb{E}(X_i), \quad \text{Var}\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j).$$

If $X_1, \dots, X_n$ are independent, every covariance term is zero.

- Induction on n from Theorems 4.57 and 4.73.

Sums of i.i.d. variables

Proposition 4.80

Let $X_1, \dots, X_n$ be i.i.d., each with mean μ , variance σ^2 and MGF M_X , and let $S = X_1 + \dots + X_n$. Then $\mathbb{E}(S) = n\mu$, $\text{Var}(S) = n\sigma^2$, $M_S(t) = M_X(t)^n$.

- Section 5 builds the binomial from n Bernoulli variables with the MGF part.
- Section 7 divides S by n and studies the sample mean.

Example 4.81

Example 4.81

In one hour a shop serves N customers, where N takes the values 0, 1, 2 with probabilities 0.2, 0.5, 0.3. Each customer's spend has mean \$40 and standard deviation \$20. The spends are independent of each other and of N . Let T be the hour's total takings. (a) Find $\mathbb{E}(T \mid N = n)$ and $\text{Var}(T \mid N = n)$. (b) Use the law of iterated expectations to find $\mathbb{E}(T)$. (c) Find $\text{Var}(N)$. (d) Use the law of total variance to find $\text{Var}(T)$ and the standard deviation of T .

Transformations of Random Variables

Four methods for a change of variable

- **List the values** (discrete): $p_Y(y) = \sum_{x: g(x)=y} p_X(x)$. Collect the mass.
- **The CDF method**: find $F_Y(y) = P(g(X) \leq y)$, then differentiate. The first step always works. The derivative is a pdf when Y is continuous.
- **The MGF method**: find $M_Y(t)$, then find a distribution with that MGF. By uniqueness (Theorem 4.42) it is the distribution of Y .
- **The Jacobian method**: a formula gives f_Y from f_X and the derivative of the inverse map.

Example 4.82

Example 4.82

X takes $-1, 0, 1, 2$ with prob. $0.1, 0.4, 0.3, 0.2$: find the pmf of $Y = X^2$.

Monotone change of variable

Proposition 4.83 (Monotone change of variable)

Let g be strictly monotone and differentiable, with $g'(x) \neq 0$ wherever $f_X(x) > 0$, and with inverse $x = g^{-1}(y)$. Take any $y = g(x)$ with $f_X(x) > 0$. Then

$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$. For every other y , $f_Y(y) = 0$.

- When g is monotone, the CDF method gives this one-line formula.
- If g is not monotone, split the range of X into monotone pieces. Use the formula on each piece and add the results.

Example 4.84

Example 4.84

X has the pdf $f_X(x) = \frac{1}{5}$ on $-3 \leq x \leq 2$, and 0 elsewhere. Let $Y = X^2$. (a) Find the pdf of Y with the CDF method. (b) Find it again with Proposition 4.83, used on two pieces. (c) Check that the pdf of Y integrates to 1.

Example 4.85

Example 4.85

X has the pdf $f_X(x) = e^{-x}$ for $x > 0$, and 0 elsewhere. Let $Y = 2X$. Find the pdf of Y in three ways: (a) with the CDF method, (b) with Proposition 4.83, (c) with the MGF method.

Example 4.86

Example 4.86

X has the pdf $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ for all real x . Take as given that f_X is a pdf. Let $Y = X^2$. (a) Find the pdf of Y with the CDF method. (b) Find it again with Proposition 4.83, used on two pieces.

Two variables: the area-scaling factor

Change of variables in two dimensions

For $(X, Y) \rightarrow (U, V)$ smooth and invertible, $f_{U,V}(u, v) = f_{X,Y}(x, y) |J|$, where

$$J = \det \frac{\partial(x, y)}{\partial(u, v)}.$$

- Here J is the Jacobian (雅可比行列式), and $|J|$ converts area in the (u, v) plane into area in the (x, y) plane.
- Four steps: solve for x and y ; compute J ; write $f_{X,Y}(x, y) |J|$; find the region where it is positive.
- A pdf is not complete without its region. For the pdf of U alone, integrate out v at the end.

Example 4.87

Example 4.87

X and Y are independent, each with the pdf $f(x) = 1$ on $(0, 1)$. So $f_{X,Y}(x, y) = 1$ on the unit square. Let $U = X + Y$ and $V = X - Y$. (a) Find the joint pdf of (U, V) and the region where it is positive. (b) Find the pdf of U . (c) Check (b) with the CDF method.

Example 4.88

Example 4.88

X_1 and X_2 have the joint pdf $f(x_1, x_2) = 4x_1x_2$ for $0 < x_1 < 1$ and $0 < x_2 < 1$, and 0 elsewhere. Let $Y_1 = X_1/X_2$ and $Y_2 = X_1X_2$. (a) Find the joint pdf of (Y_1, Y_2) and the region where it is positive. (b) Find the pdf of Y_1 . (c) Find the pdf of Y_2 . (d) Check the pdf of Y_1 on $0 < y_1 \leq 1$ with the CDF method.